Data-driven Adaptive Model Predictive Control for Wind Farms: A Koopman-Based Online Learning Approach

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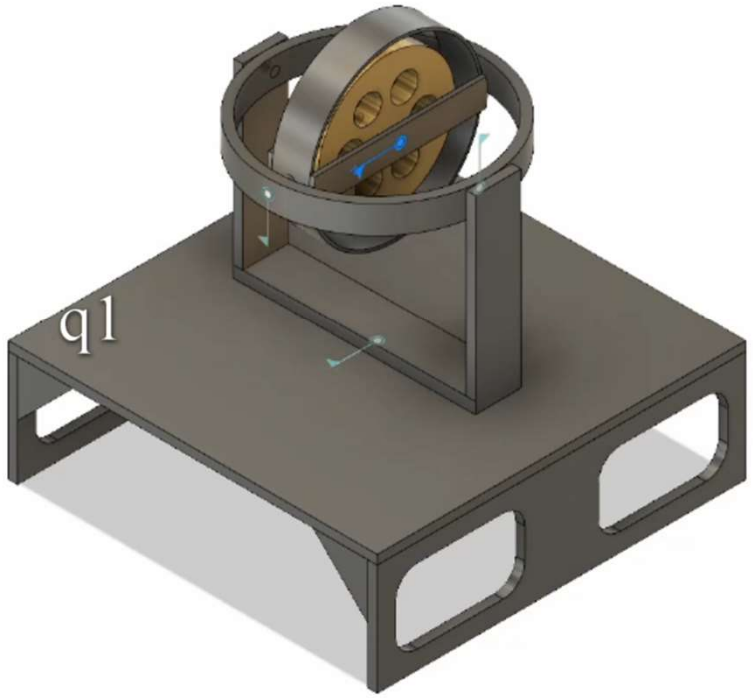
Motivation
Koopman
Adaptive MPC
Results
Summary

Koopman Models for Adaptive Model Predictive Control

- Koopman theory for model predictive control (MPC)
  - Model non-linear physics as linear state space models
  - Linearized models in quasi linear MPC (qLMPC)

- MPC performance depends on model accuracy
  - Parameter uncertainties / changes
  - Unknown disturbances

- Previous work adaptive Koopman MPC:
  - PoC on Gyroscope [Cisneros et al. 2021]
  - Forgetting factor [Calderón et al. 2021]



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Wind Farm Control

- Wind farms
  - Maximize possible yield per available area, minimize operation and maintenance cost
  - Wake interaction result in power losses

- Wind farm control objectives
  - Minimize error to reference power / maximize generated power
  - Minimize damage equivalent loads
  - Minimize changes in control input

- Control strategies
  - Wake redirection control
  - Axial induction control





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Koopman Framework for Wake and Wind Farm Models

- Koopman MPC [Cassamo et al. 2021]
  - 100,000 grid points of 3D wind field (SOWFA) → Dynamic mode decomposition and singular value decomposition for state reduction
  - Flattened 3D structure to vectors → appended vectors for Koopman system identification

- Realtime Koopman MPC [Sharan et al. 2022]
  - Improved power reference tracking with physically (Navier-Stokes equation) motivated Koopman models
  - Real time applicability:
    - Identified Koopman wake models based on average wind speed at rotor disc (vs. 100.000 grid points)
    - Reformulated qLMPC as quadratic programming problem

- Open questions
  - Does the non-adaptive qLMPC, designed for steady-state wind at constant speed, work for changing wind speeds?
  - Can we apply the adaptive Koopman MPC from [Cisneros et al. 2021] without changes?
  - Which changes are necessary in our wind farm simulation for performant reference power tracking?



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Koopman: Overview and Example

- Koopman operator:
  - Nonlinear finite dimensional → linear infinite dimensional system
  - Goal: Finite-dimensional approximation of Koopman operator

- Example [Brunton et al. 2016]: Second order non-linear model

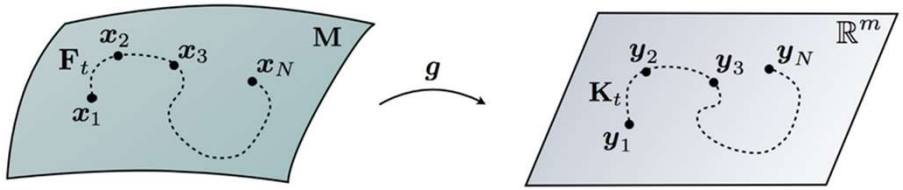
$$x_1(k+1) = \mu x_1(k)$$
$$x_2(k+1) = \lambda x_2(k) - \lambda x_1^2(k) + u(k)$$

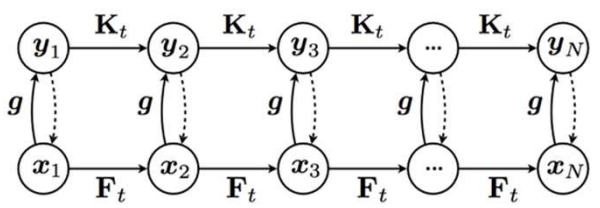
- Introducing new variable

$$g(k) = x_1^2(k) \implies g(k+1) = x_1^2(k+1) = \mu^2 x_1^2(k) = \mu^2 g(k)$$

- Third order linear model

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ g(k+1) \end{bmatrix} = \begin{bmatrix} \mu & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & \mu^2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ g(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u(k)$$





$\mathbf{F}_t$: $x_k \mapsto x_{k+1}$

g : $x_k \mapsto y_k$

$\mathbf{K}_t$: $y_k \mapsto y_{k+1}$

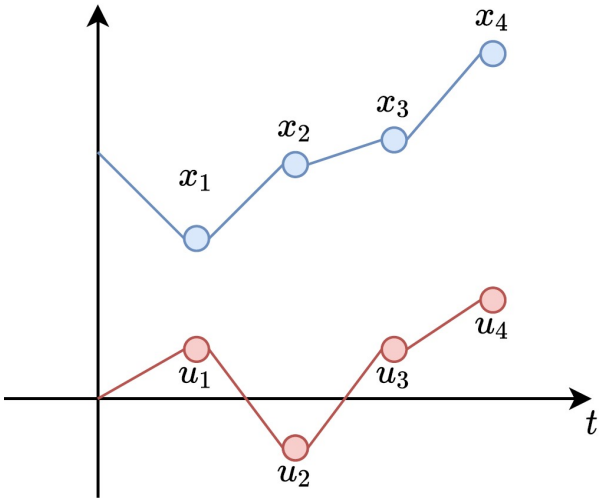
Koopman: Approximation of the Koopman Operator

Use extended dynamic mode decomposition (EDMD) algorithm [Calderón et al. 2021]

Requirements:

- Data:
  - $X = [x_1 \quad x_2 \quad \cdots \quad x_{N-1}]: \mathbb{R}^{n_x \times n_{N-1}}$
  - $X_+ = [x_2 \quad x_3 \quad \cdots \quad x_N]: \mathbb{R}^{n_x \times n_{N-1}}$
  - $U = [u_1 \quad u_2 \quad \cdots \quad u_{N-1}]: \mathbb{R}^{n_u \times n_{N-1}}$
- Lifting functions

$$\Phi_k^\top = \left[x_k^\top \; g(x_k)^\top\right] : \mathbb{R}^{n_x} \times \mathbb{R}^m \to \mathbb{R}^{n_\varphi}$$



Lift the data:
$G = \Phi(X)$
$G_+ = \Phi(X_+)$

→

Formulate:
$\min_{A,B} ||G_+ - AG - BU||_F$

→

Solve:
$K = [A, B] = G_+ \begin{bmatrix} G \\ U \end{bmatrix}^\dagger = L\, L_u^\dagger$

Adaptive MPC: Update Rule

- Mismatch between past predictions and data → update
- Store past predictions and data vector $[z_{k-5}, z_{k-4}, \dots z_k]$
- Check update condition

$$\left\| \begin{bmatrix} x_{k-h} \\ x_{k-h+1} \\ \vdots \\ x_k \end{bmatrix} - \begin{bmatrix} C\hat{K}\phi(z_{k-h-1}) \\ C\hat{K}\phi\left(\begin{bmatrix} \hat{K}\phi(z_{k-h-1}) \\ u_{k-h} \end{bmatrix}\right) \\ \vdots \\ C\hat{K}\phi\left(\begin{bmatrix} \hat{K}\phi\left(\begin{bmatrix} \hat{K}\phi(\dots) \\ u_{k-2} \end{bmatrix}\right) \\ u_{k-1} \end{bmatrix}\right) \end{bmatrix} \right\|_\infty \geq \nu$$

- Update P, L_u and L

$$P^+ = P + 1$$

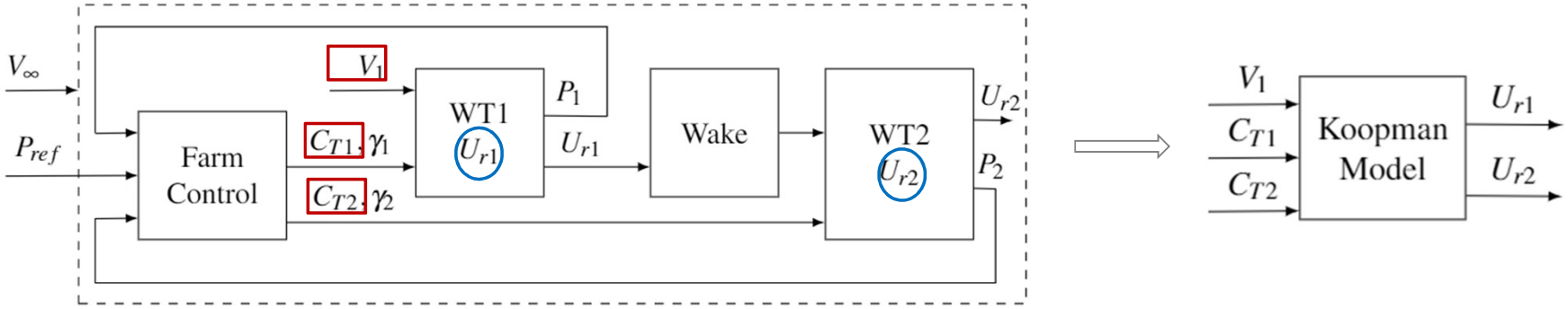
$$L_u^+ = \frac{1}{P^+}\left((P^+ - 1)L_u + \phi(z_{k,5})\phi(z_{k,5})^T\right)$$

$$L^+ = \frac{1}{P^+}\left((P^+ - 1)L + \phi(z_{k,5}^+)\phi(z_{k,5})^T\right)$$

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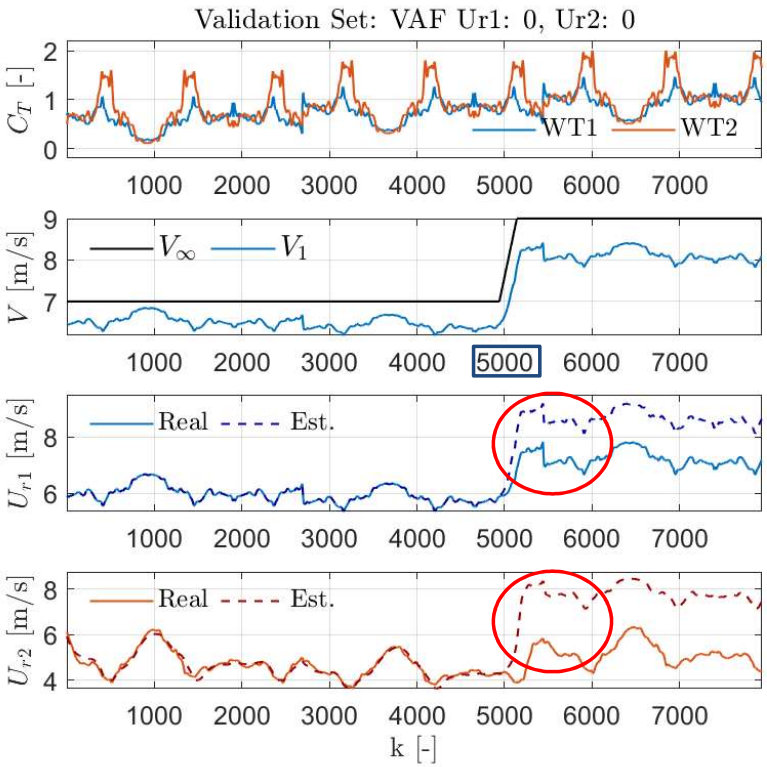
Adaptive MPC: Wind Farm Simulation

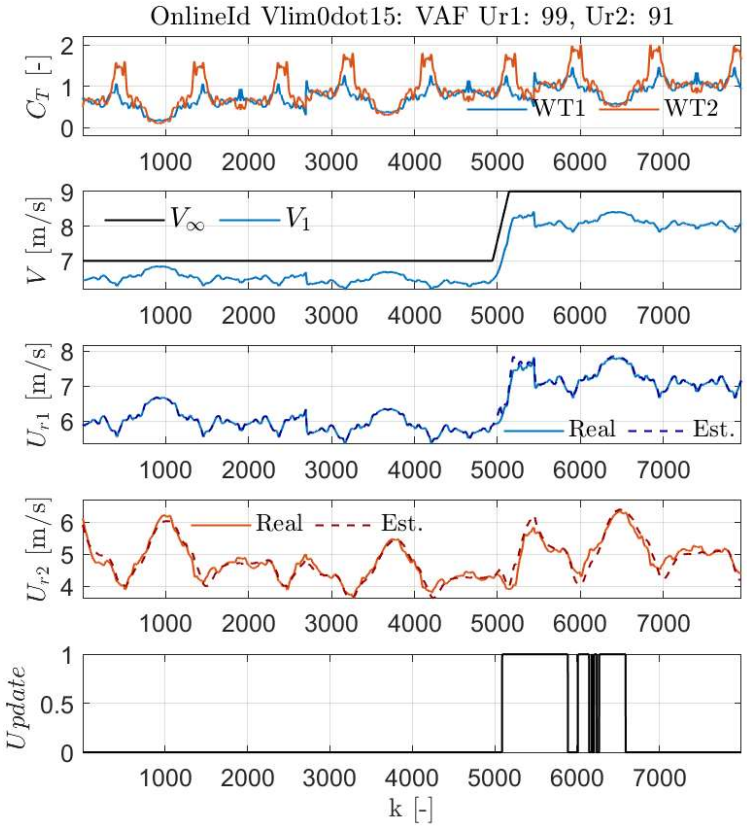
- WFSim [Boersma et al. 2016]: Open loop data acquisition and closed loop controller tests
- [Sharan et al. 2022]: Non-adaptive Koopman MPC with constant free flow wind $V_\infty = 8$ m/s
  - Input: Thrust control signals C_{T1} and C_{T2},
  - Outputs: Average wind speeds acting on rotor discs U_{r1} and U_{r2}
- Adaptive Koopman: Initial model with inputs: Thrust C_{T1}, C_{T2}, and speed V_1



- Collected data for constant wind V_∞ at 7 m/s and wind change V_∞ from 7 to 9 m/s

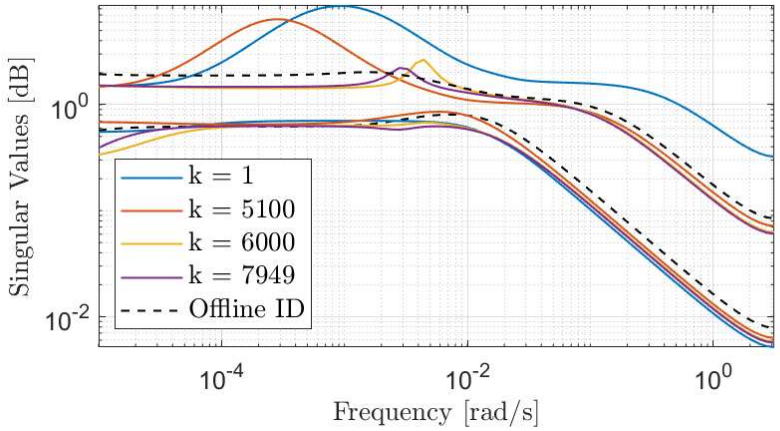
Open Loop System Identification: Offline vs. Online



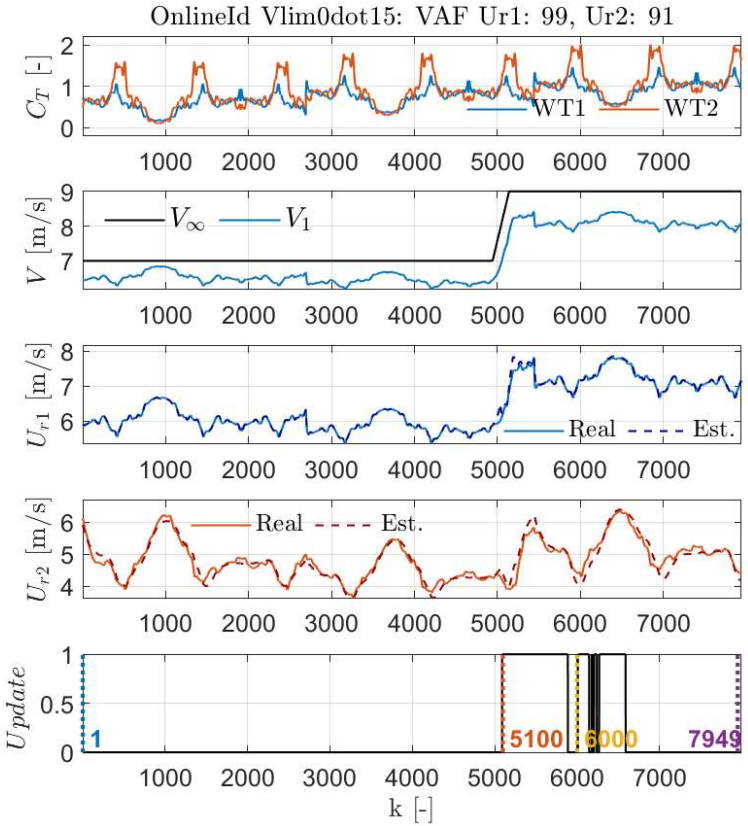


- Good trajectory following up to chance from 7 to 9 m/s, fails at wind speed change
- Variance-accounted-for adaptive Koopman: VAF > 90%

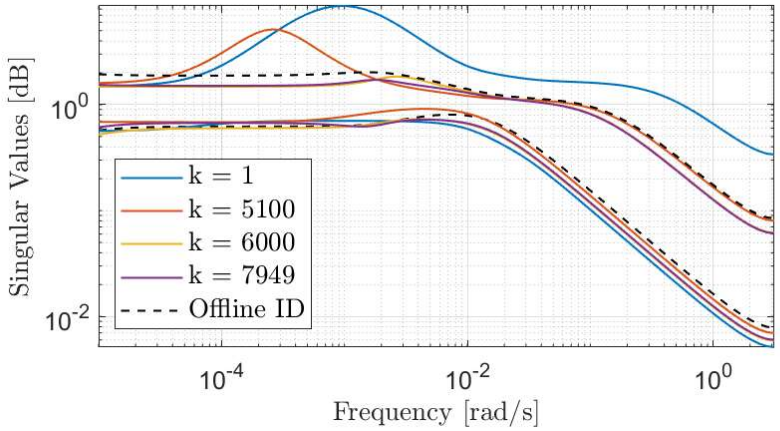
System Identification Evaluation: Singular Values (η = 0.15 m/s)

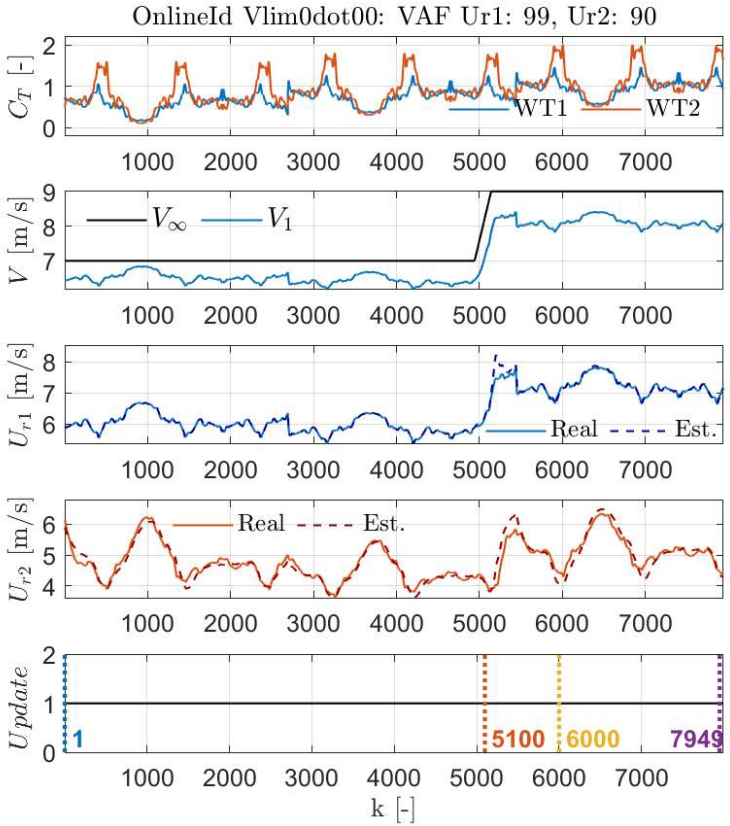


- Singular values of linear Koopman models at sample times **1**, **5100**, **6000** and **7949**
- η = 0.15 m/s: No update during constant wind speed, update starts at wind step change
- Samples k > 6700: Difference measured to estimated wind speed < 0.15 m/s



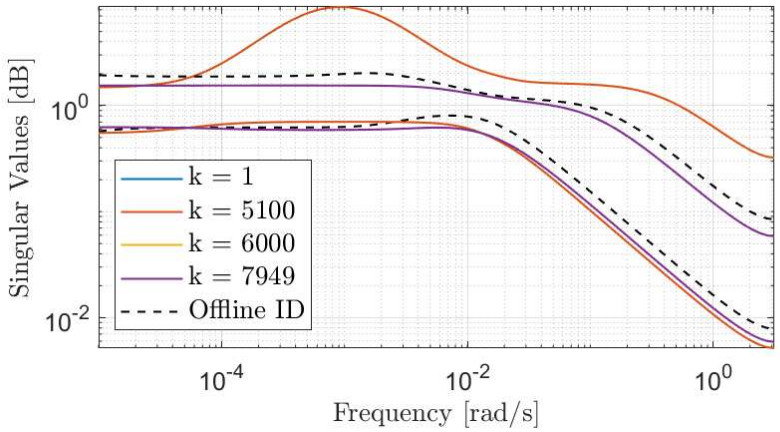
System Identification Evaluation: Singular Values (η = 0 m/s)



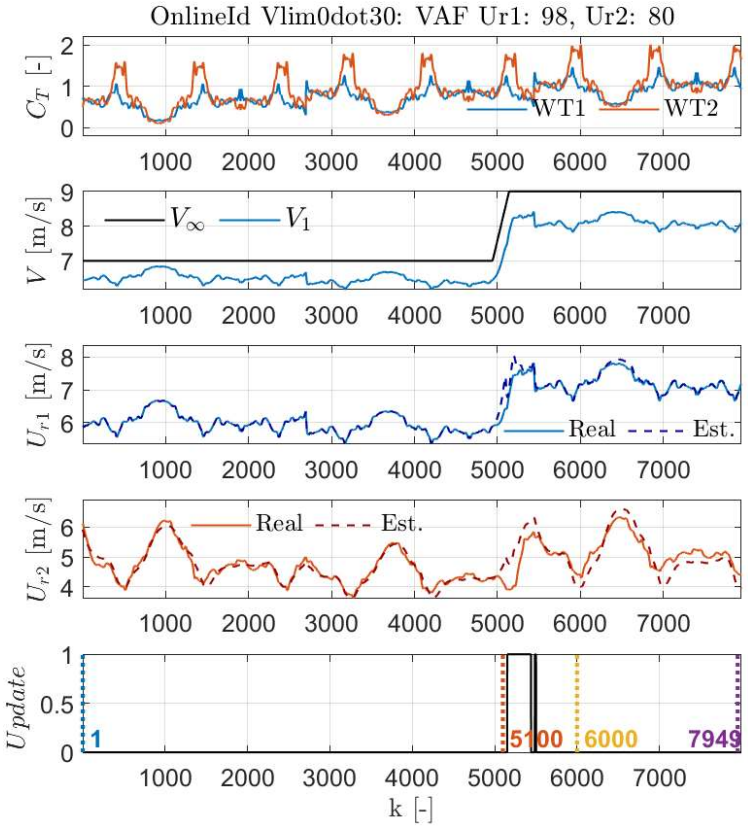


- Singular values of linear Koopman models at sample times **1**, **5100**, **6000** and **7949**
- v = 0 m/s: Update at every step
- Comparison 0 to 0.15 m/s
  - More update calculations
  - Negligible improvement in SV comparison
  - No improvement regarding VAF

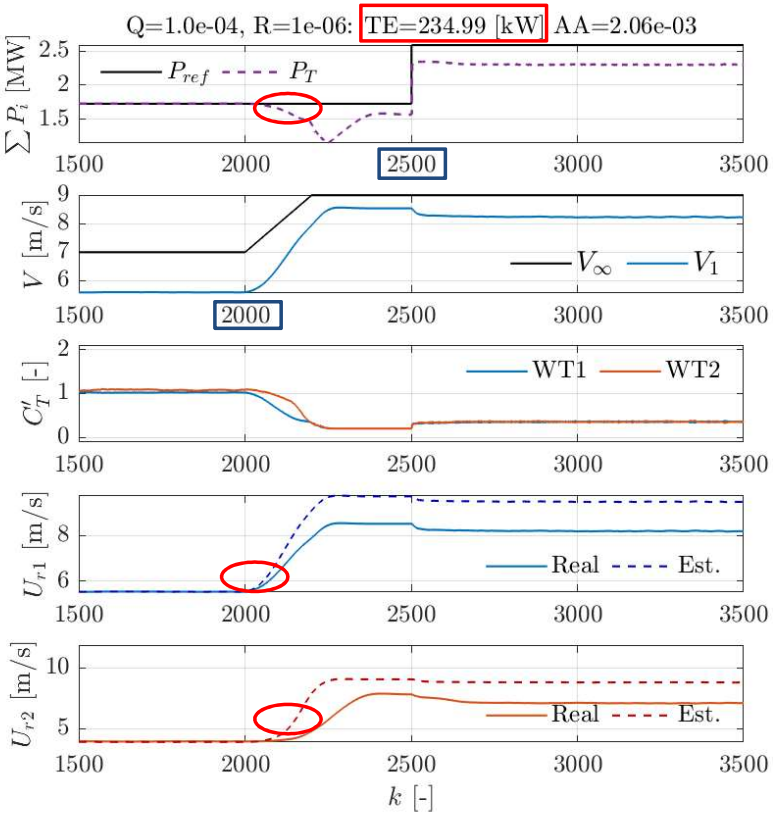
System Identification Evaluation: Singular Values (η = 0.3 m/s)

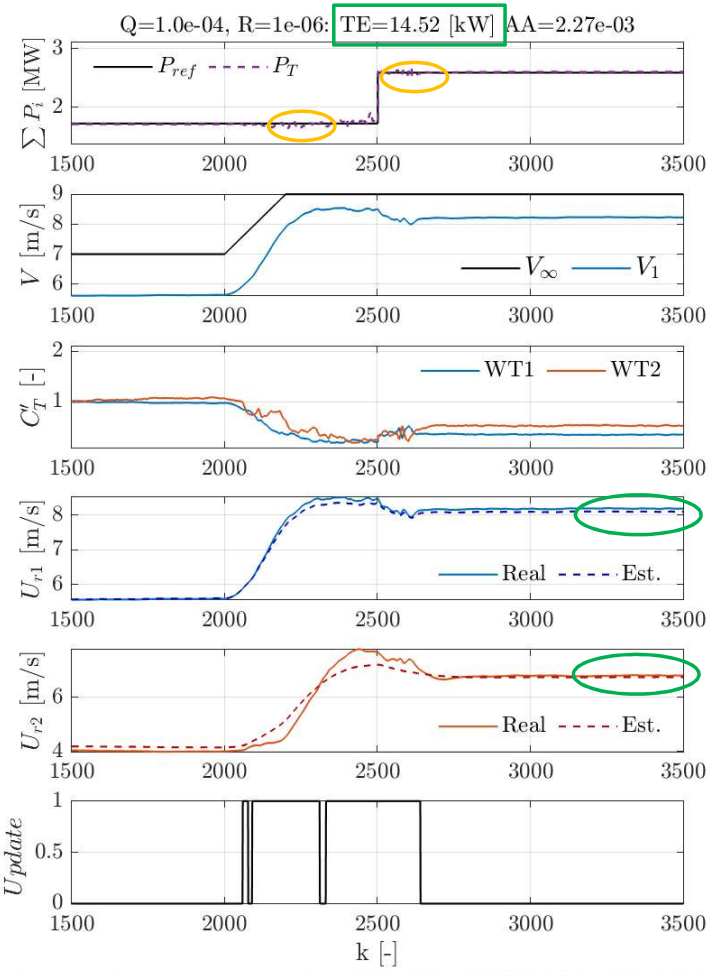


- Singular values of linear Koopman models at sample time 1 (= 5100) and 6000 (= 7949)
- Comparison 0.3 to 0.15 m/s
  - Less update calculations
  - Degradation in SV comparison
  - Degradation regarding VAF (80 vs. 91 %)



Adaptive Koopman MPC: Closed Loop



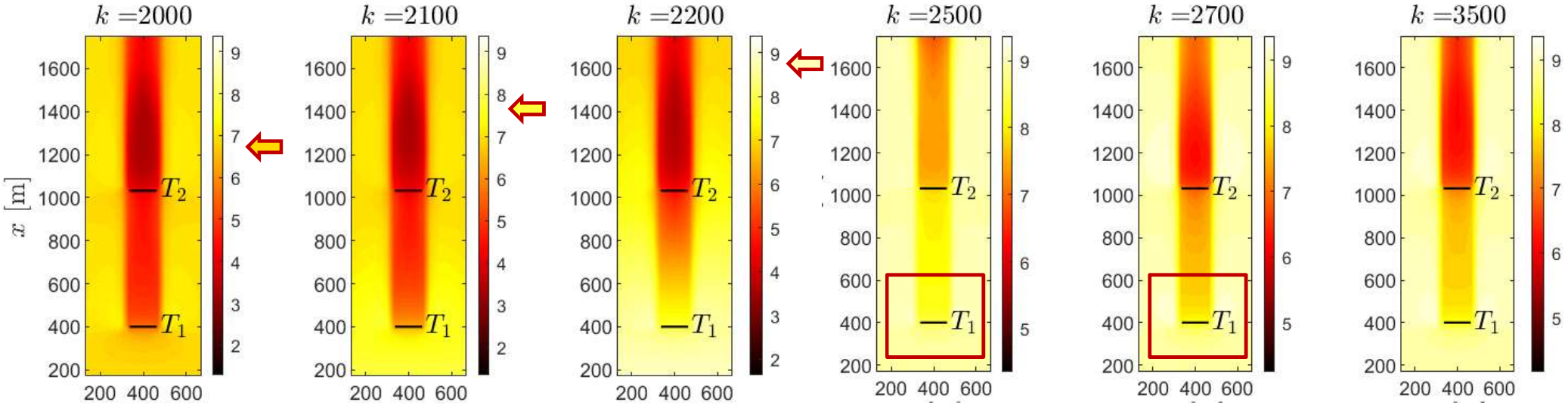


- Testsignal: Pref 1.72 → 2.58 MW, Vinf 7→ 9 at k = 2000
- Non-adative: Offset in wind estimates → power offset

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Adaptive Koopman MPC: Wind Field



V_∞ = 7 m/s

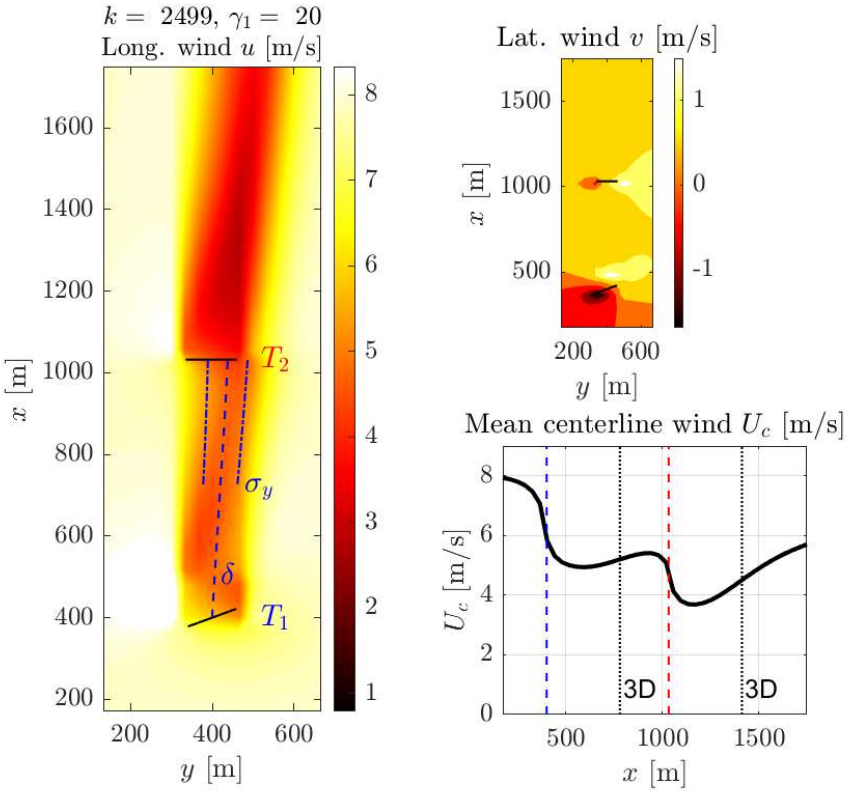
V_∞ ca. 8 m/s

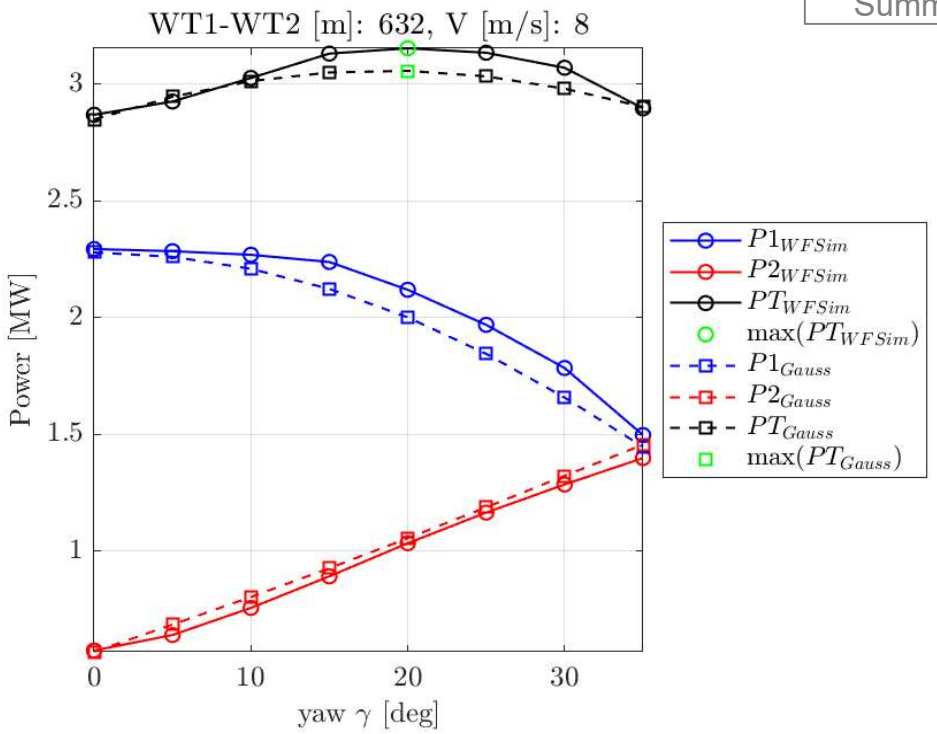
V_∞ = 9 m/s

- Test signal: k = 2000: V_∞ 7→ 9 m/s to 9 m/s
- k = 2000 to k = 2200: Wind V_∞ increase

- Test signal: k = 2500: Pref 1.72 → 2.58 MW
- k= 2500: Reference power increase: C_{T1} and C_{T2} increase, wind in wakes decreases

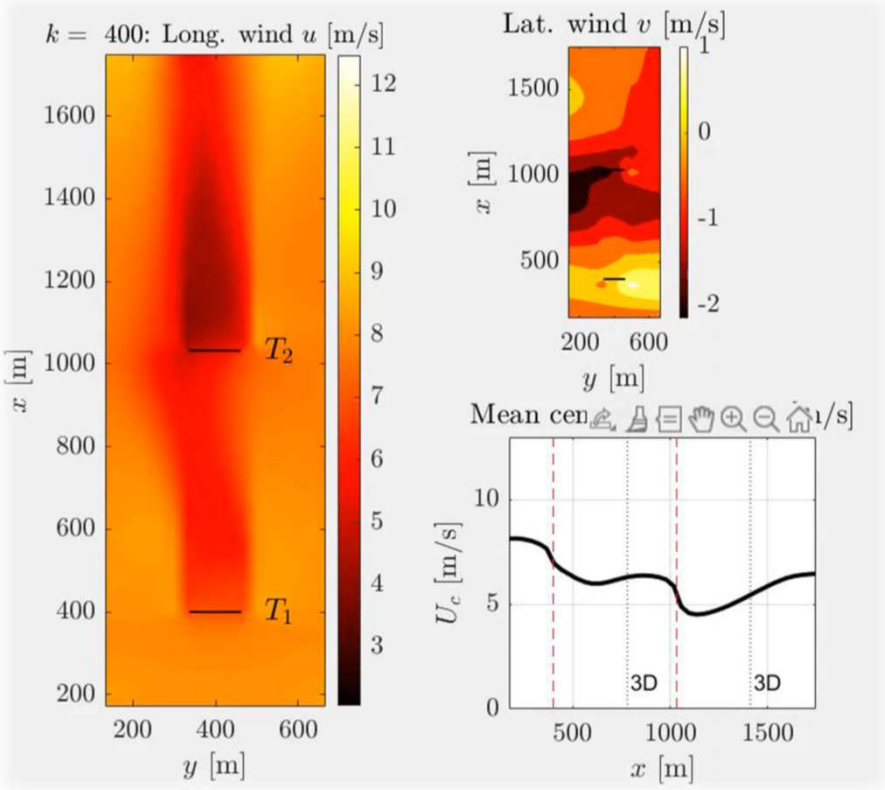
Outlook: Combined Axial Induction/Wake Redirection Control





- Additional wake redirection control via yaw angle
- Open loop simulation: Power increases up to 10%

Outlook: Turbulent wind



- Use more realistic wind: Update boundary conditions for both wind speed and direction

Summary and Outlook

- Adaptive Koopman MPC for wind farm control with axial induction control:
  - Update rule is presented for online learning
  - Parameter fine tuning in open loop
  - Good closed loop performance (with small additional excitation signal)

- Current work
  - Combination of axial induction and yaw control (non-adaptive at constant wind, submitted to IFAC)
  - Test cases with turbulent wind

- Future work:
  - Adaptive controller for combined thrust and yaw control
    - Automate meta-parameter tuning
    - Leverage forgetting factor
  - Simulation in medium and high fidelity simulation environment



Thank you for your attention!

